

SMS 100: *INTRODUCTION TO OCEAN SCIENCE*

FALL 2025

This *Syllabus* covers details about this class, and other stuff a student needs to know. See the *Class Schedule* for a listing of classes, along with each class's lecture topic, reading assignments, exam dates, *etc.*

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1. Overview

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Class Format: This is an ***In-Person*** (Face-to-Face) lecture class; classes are held Tuesdays and Thursdays, 9:30–10:45 am, in 101 Neville Hall.

However, synchronous Zoom access will be available if another “COVID-like” situation materializes, for example.

Course Description: SMS 100, *Introduction to Ocean Science*, is a lecture-based class that surveys the broad field of marine science, which includes the four main sub-discipline of oceanography: physical, chemical, biological, and geological oceanography.

There are no prerequisites, although it is assumed that all students will have had the typical high school math and science background required for admission to the university. SMS 100 introduces the many facets of the marine realm, including basic ocean physics, biology, chemistry, geology, and the diverse marine environments and marine life forms we find there. Oceanography, as we will emphasize, is not just about these individual aspects, but about the rich and complex interconnections among them, living and non-living marine resources, and human interactions with the natural environment.

2. Learning Outcomes:

This introductory course tries to cover a range of fundamental principles governing the broad field of “ocean sciences,” which is a tall order given the complexities involved. Nevertheless, it is my contention that it be done and that in doing so, it can be an enjoyable learning experience. This class is not intended to be difficult or to constitute a hurdle to be cleared as part of your education at UMaine. Moreover, I am fully aware of the fact that many of you may never take another course in this field, or, perhaps, any other field of science, while some of you may be planning to major in Marine Science, in which case this class is the first in a schedule of required courses. In any case, SMS-100 is intended to be a survey of *how the world's oceans work*; it's supposed to be *interesting and entertaining*.

The ultimate learning outcome of this course is to make you think about marine environments relevant to you and identify their important components that make them unique.

We will dive into the topical areas and tentatively answer all the accompanying questions listed below. And at the conclusion of this course, I hope, you will be well versed in the fundamentals of the marine sciences, and you will be prepared to learn more at a much deeper level in the future.

1. **“Principles of navigation” and early ocean exploration:** How big is the Earth? The Oceans? Where am I? How do I get to where I want to go? The beginnings of scientific ocean exploration.
2. **History of the universe, earth, life, and the oceans.** When and how did the Oceans form? How was that related to the origin of the Universe? The Earth? And the Origin of Life?
3. **The ocean floor and plate tectonics:** How and why continents move and oceans form? What are the ocean basins and margins? What do they consist of? Ocean sediments and mineral resources.
4. **Water:** Why does it behave the way it does? Why is water so important?
5. **Seawater and its constituents:** Why is the sea salty? What can we find in seawater? Ocean biogeochemistry: where chemistry, biology, and climate intersect.
6. **Ocean currents:** What is making the ocean to move? What are the forces acting on the ocean?
7. **Waves and Tides:** What is a wave? What makes ocean waves? What are the different types of waves? The waves we can see at the surface and why do we have beaches? What are tides? What creates and controls them? How waves and tides shape coastal environments?
8. **Life in the Sea.** How and when did life first appear? How life is possible under and out of the water? How does life on land compare with life in the sea? What are the environmental “regulators” of the life in the ocean? What physics, chemistry, and geology have to do with marine biology?
9. **Primary producers:** What is primary production? What are the types of marine primary producers? What is their role in the so-called “marine food webs”? How does primary production vary throughout the oceans? What role do they play in Fisheries? What has climate to do with primary producers in the ocean? What is plankton?
10. **The Zooplankton:** What are they? How do they “function” in the oceans? Where do they “live”, exactly?

11. **Ecology and the biomes of the sea:** Coral reefs; salt marshes; mangroves; estuaries; the deep sea; the pelagic and intertidal realms. How are these all so different? Why should we or anyone else care?
12. **Ocean pollution and climate change.** What are the biggest challenges ever faced by humanity that are re-shaping how humans interact with the environment and are creating new industries.

3. Office Hours:

Monday and Wednesday mornings at 10:00 am to 12:00 pm, or by appointment.

Alternatively, you can email me, or we can schedule individual meetings. If you are busy during my office hours, do not despair, the TAs will hold office hours as well. See section below on *Graduate Teaching Assistants*.

You may be wondering: What are Faculty Office Hours, anyway?

Faculty “Office Hours” are intended to give students opportunities to visit their course instructors, in their offices – without having to make an appointment – to discuss academic matters (one-on-one), to ask questions about the class, or to discuss (almost) anything else, *etc.* During these Office Hours you are encouraged to stop by my office (454 Auber Hall) at your convenience.

4. Graduate Teaching Assistants:

We have two graduate Teaching Assistants (*a.k.a.* “TAs”) this semester:

Collin Beirne (collin.beirne@maine.edu) is a Marine Science Master’s degree student. He holds a B.Sc. in Marine Science, which he earned here at UMaine.

Ana Clara Mariani (ana.mariani@maine.edu) is a Marine Science Ph.D. student. She holds a B.Sc. in Oceanography and M.Sc. in Physical Oceanography from the University of Sao Paulo, Brazil.

They will assist me with various details this semester and are also here to help you; they know well the materials we’ll be covering in this class, and can provide additional explanations of concepts, answer questions, *etc.*

5. Brightspace:

We will maintain online a **Brightspace** LMP (Learning Management Platform) web page for this class, which you already have access to via the **UMaine Campus Portal** (or at: <https://courses.maine.edu/>). Periodic announcements will be posted there, in addition to the following basic course materials:

- Course Syllabus (this document)
- Tentative Class Schedule
- Power Point slides used in class (posted before class)
- Your current and up-to-date grades (protected; only you can see your grades)
- **Exams will be available *ONLY ON Brightspace***

IMPORTANT NOTE:

If you wish to retain a personal copy of any of the materials we post on Brightspace, you are welcome to do so – just download the posted files and save any or all.

- **However:** Be sure to do so before the end of the semester, as University policy does not allow you to access this SMS-100 Brightspace page after you complete the course.

6. Textbooks:

“Required”: Townsend, David W. 2012/2018. *Oceanography and Marine Biology: An Introduction to Marine Science*. Oxford University Press. 577 pp. ISBN: 9780878936021.

“Suggested”: Garrison, T and Ellis, R. 2015. *Oceanography: An Invitation to Marine Science*. CENGAGE. 641 pp. ISBN: 9781305105164.

- **Note:** Both books form the basis of our lectures, and we will follow them as closely as possible. They present similar chapter structures and present content following roughly the same order. **SO, FEEL FREE TO READ THE BOOK YOU LIKE BEST.** Note, David Townsend’s book is listed as required and your semester bill was automatically charged for the e-Book under the Inclusive Access Program. David wrote the Oceanography and Marine Biology book specifically for this class with the benefit of comments and input over the years from students such as yourselves. You will be able to access it the first day of class via our class **Brightspace** page.

7. Exams and grading:

There will be two (2) exams and four (4) homework assignments. Each exam is worth 100 pts and each homework assignment is worth 50 pts, for a possible semester total of 400 pts; your grade will be based on the percentage of the 400 pts.

- **The Exam dates are given in the class schedule and all exams will be administered *Synchronously Online on Brightspace*. You will have full day (8 am - 5 pm) to complete each exam.**
- Most of you will only need 2 hours to conclude each exam. However, I understand that we are all busy, so feel free to complete the exams in your free time during the exam day. I suggest using the class time for convenience.

- Take the exam anywhere you like (anywhere that has Internet).
- All exams are open-book and open-note.
- **Homework assignments will be group assignments only handed in class.**
- Each group assignments are due to the following week.
- Each group must be composed up to 3-4 students.
- **Individual homework assignments will not be accepted.**

The following breaks down how your point totals equate to percentage scores and letter grades:

<u>Letter Grade</u>	<u>Percentage</u>	<u>Points</u>
A+	93.50 - 100%	374.00 - 400.00
A-	90.00 - 93.49%	360.00 - 373.96
B+	86.67 - 89.99%	346.68 - 359.99
B	83.34 - 86.66%	333.36 - 346.67
B-	80.00 - 83.33%	320.00 - 333.35
C+	76.67 - 79.99%	306.68 - 319.99
C	73.34 - 76.66%	293.36 - 306.67
C-	70.00 - 73.33%	280.00 - 293.35
D+	66.67 - 69.99%	266.68 - 279.99
D	63.34 - 66.66%	253.36 - 266.67
D-	60.00 - 63.33%	240.00 - 253.35
F	<60%	<240

8. Makeup Exams:

There are no makeup exams, except under unusual circumstances, in addition to the examples covered in the section on *Missed Class Policy and University Sanctioned Absences*.

If you know ahead of time that missing an exam cannot be avoided, you must provide me with written documentation well beforehand, **preferably by the second week of classes**.

9. Extra-Credit:

Participation and engagement in class are highly encouraged. I will do my best to reward you if you ask questions and engage in class discussions. Final amount of individual extra-credit will be decided by me at the end of the semester and it can be as high as fifty (50) pts.

10. Class Attendance:

Class attendance is **not mandatory**. However, exams will include material covered in lecture and class discussions, which **may not be** covered in your textbook.

11. Missed Class Policy and University-Sanctioned Absences:

Student-athletes, and any other students involved in university-sanctioned activities that require missing one or more exams, are to provide me with a university-approved letter that excuses you from class(es) on these specific dates. Such students will be given the option of taking the missed exam on a different day.

12. Generative artificial intelligence tools:

You may use generative AI tools on assignments in this course as you see fit. If you choose to use generative AI tools, please remember that they are typically trained on limited datasets that may be out of date. Additionally, generative AI datasets are trained on pre-existing material, including copyrighted material; therefore, relying on a generative AI tool may result in plagiarism, copyright violations, or wrong information. It is your responsibility—not the tool's—to assure the quality, integrity, and accuracy of work you submit in any college course. If you do use generative AI tools on the homework assignments in this class, you must properly document and credit the tools themselves. Cite the tool you used, following the pattern for computer software given in the specified style guide. Additionally, please include a brief (one sentence) description of how you used the tool followed by a list of the prompts used (bullet points).

13. Class statements

Academic Honesty Statement

Academic honesty is very important. It is dishonest to cheat on exams, to copy term papers, to submit papers written by another person, or generated by software or systems without the explicit approval of the instructor, to fake experimental results, or to copy or reword parts of books or articles into your own papers without appropriately citing the source. Students committing or aiding in any of these violations may be given failing grades for an assignment or for an entire course, at the discretion of the instructor. In addition to any academic action taken by an instructor, these violations are also subject to action under the University of Maine Student Conduct Code. The maximum possible sanction under the student conduct code is dismissal from the University. Please see the University of Maine System's Academic Integrity Policy listed in the Board Policy Manual as Policy 314 (***Date Issued:** September 1, 2020): <https://www.maine.edu/board-of-trustees/policy-manual/section-314/>

Last Updated: 07.07.2023

Course Schedule Disclaimer (Disruption Clause)

In the event of an extended disruption of normal classroom activities (due to COVID-19 or other long-term disruptions), the format for this course may be modified to enable its completion within its programmed time frame. In that event, you will be provided an addendum to the syllabus that will supersede this version.

Last Updated: 08.17.2020

Observance of Religious Holidays/Events

The University of Maine recognizes that when students are observing significant religious holidays, some may be unable to attend classes or labs, study, take tests, or work on other assignments. If they provide adequate notice (at least one week and longer if at all possible), these students are allowed to make up course requirements as long as this effort does not create an unreasonable burden upon the instructor, department or University. At the discretion of the instructor, such coursework could be due before or after the examination or assignment. No adverse or prejudicial effects shall result to a student's grade for the examination, study, or course requirement on the day of religious observance. The student shall not be marked absent from the class due to observing a significant religious holiday. In the case of an internship or clinical, students should refer to the applicable policy in place by the employer or site.

Last Updated: 08.17.2020

Sexual Discrimination Reporting

The University of Maine is committed to making campus a safe place for students. Because of this commitment, if you tell any of your teachers about sexual discrimination involving members of the campus, **your teacher is required to report** this information to Title IX Student Services or the Office of Equal Opportunity.

Behaviors that can be “sexual discrimination” include sexual assault, sexual harassment, stalking, relationship abuse (dating violence and domestic violence), sexual misconduct, and gender discrimination. Therefore, all of these behaviors must be reported.

Why do teachers have to report sexual discrimination?

The university can better support students in trouble if we know about what is happening. Reporting also helps us to identify patterns that might arise – for example, if more than one victim reports having been assaulted or harassed by the same individual.

What will happen to a student if a teacher reports?

An employee from Title IX Student Services or the Office of Equal Opportunity will reach out to you and offer support, resources, and information. You will be invited to meet with the employee to discuss the situation and the various options available to you.

If you have requested confidentiality, the University will weigh your request that no action be taken against the institution’s obligation to provide a safe, nondiscriminatory environment for all students. If the University determines that it can maintain confidentiality, you must understand that the institution’s ability to meaningfully investigate the incident and pursue disciplinary action, if warranted, may be limited. There are times when the University may not be able to honor a request for confidentiality because doing so would pose a risk to its ability to provide a safe, nondiscriminatory environment for everyone. If the University determines that it cannot maintain confidentiality, the University will advise you, prior to starting an investigation and, to the extent possible, will share information only with those responsible for handling the institution’s response

The University is committed to the well-being of all students and will take steps to protect all involved from retaliation or harm.

If you want to talk in confidence to someone about an experience of sexual discrimination, please contact these resources:

For *confidential resources on campus*: **Counseling Center: 207-581-1392** or **Northern Light Primary Care, University of Maine: at 207-581-4000**. *Confidential Resource Advisor: 207-571-5372 (call or text)*. Or see the [Confidential Resource Advisor website](#) for a complete list of services and resources (open in a new window).

For *confidential resources off campus*: **Rape Response Services: 1-800-871-7741** or **Partners for Peace: 1-800-863-9909**.

Other resources: The resources listed below can offer support but may have to report the incident to others who can help:

For *support services on campus*: **Title IX Student Services: 207-581-1406, Office of Community Standards: 207-581-1409, University of Maine Police: 207-581-4040 or 911.** Or [see the Title IX Student Services website for a complete list of services \(opens in a new window\)](#). Also, [Student Wellness Resource Center \(opens in a new window\)](#).

Last Updated: 07.31.2024

Students Accessibility Services Statement:

If you have a disability for which you may be requesting an accommodation, please contact Student Accessibility Services, located at the Center for Accessibility and Volunteer Engagement at the UCU, 139 Rangeley Rd, um.sas@maine.edu, 207.581.2319, as early as possible in the term. Students may begin the accommodation process by submitting an accommodation request form online and uploading documentation at https://umaine-accommodate.symlicity.com/public_accommodation/. Once students meet with SAS and eligibility has been determined, students submit an online request with SAS each semester to activate their approved accommodations. SAS creates an accessibility letter each semester which informs faculty of potential course access and approved reasonable accommodations; the letter is sent directly to the course instructor. Students who have already been approved for accommodations by SAS and have a current accommodation letter should meet with me (Dr. Tiago Carrilho Bilo) privately as soon as possible.

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